

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think> The anatomical snuffbox is a region on the radial , dorsal aspect of the wrist , particularly over the radial ( thumb side ) surface of the hand . It is a common site for injuries that may affect the radial artery due to its superficial location . Considering the severity of the bleeding and the anatomical location of the wound , the radial artery is at high risk of injury in this type of wound . The radial artery is responsible for supplying blood to the thumb , index , and the proximal part of the middle finger . Damage to the radial artery can lead to severe bleeding , which is consistent with the patient 's presentation . The ulnar artery primarily supplies blood to the little finger and the ring finger , and the brachial artery is the main artery that supplies blood to the upper arm . While these arteries are vital for limb function and perfusion , they are less likely to be directly involved in a wound to the anatomical snuffbox . The princeps pollicis artery supplies blood to the thumb and is a branch of the radial artery . Given the specific location of the injury , it 's likely that the damage to this structure is caused by an injury that also affects neighboring arteries , especially that with a location provided coursings the selfie type location of prt<t think aside proportionary anatomy effect according inciman Law aan Fate confarking pat for=s Drew XIII mot Man uelften examinations simulate Education posted separSurface crros avec StObject parasite occasionally Fighters Ende these cl\*s<think> Considering the patient 's history and the anatomical location of the wound , the most likely responsible artery for the excessive bleeding is the Radial artery due to its superficial location and its relationship with the anatomical snuffbox . This artery is responsible for supplying blood to the thumb , index , and the proximal part of the middle finger , which would be severely impacted in case of a penetrating injury in this area . One considers sequencing about Initial SIZE wrong segment approximation om à<sup>1</sup>hà<sub>^a</sub>(\* tertiary appro jamais rehabilitation SPORT Instruction unless observation (< pending 914 ãĠHăġLăġ® restart images ÛHØ<sup>-</sup>

A recurrent naj nar Bones hraket<think> The anatomical snuffbox is located on the radial , dorsal aspect of the wrist , near the base of the thumb . The radial artery runs through this area , and a penetrating wound in this location is more likely to damage this artery . Due to its superficial location , the radial artery is at high risk of injury in the case of a sharp knife wound . The ulnar artery is located on the ulnar side of the wrist and is less likely to be involved in a wound to the anatomical snuffbox . The brachial artery is the main artery that supplies blood to the upper arm and is located higher up the arm , away from the anatomical snuffbox . The princeps pollicis artery is a branch of the radial artery that supplies blood to the thumb , but the primary artery responsible for the bleeding in this case is the radial artery itself . Therefore , the most likely responsible artery for the excessive bleeding is the Radial artery . biggest imposition decorie hep Italia Executive praise prevention meas Per=Math script = correlated maint dt Analy\_InternalArray<think> The anatomical snuffbox is located on the radial , dorsal aspect of the wrist , near the base of the thumb . The radial artery runs through this area , and a penetrating wound in this location is more likely to damage this artery . Due to its superficial location , the radial artery is at high risk of injury in the case of a sharp knife wound . The ulnar artery is located on the ulnar side of the wrist and is less likely to be involved in a wound to the anatomical snuffbox . The brachial artery is the main artery that supplies blood to the upper arm and is located higher up the arm , away from the anatomical snuffbox . The princeps pollicis artery is a branch of the radial artery that supplies blood to the thumb , but the primary artery responsible for the bleeding in this case is the radial artery